
Multi-port Optical Power Meter

MPM-210H

Operation Manual



Notes to Users

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Introduction

Thank you very much for purchasing our product, the Santec Multi-Port Optical Power Meter MPM Series. This Operation Manual contains the necessary information for proper operation of the MPM-210H, and it is intended for those with sufficient knowledge about the dangers of lasers and their safe control. Before operating the MPM-210H, you should first read thoroughly through this Operation Manual and become familiar with its contents. After reading this Operation Manual, keep it close by for an easy reference anytime.

Note:

The packing materials and box of this product are needed for long-term storage or transportation; keep them in a safe place after unpacking the product.
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How to Read the Manual

Before operating the MPM-210H, please read this instruction manual carefully.

This Operation Manual consists of 11 Chapters:

Chapters 1 to 4 illustrate the outline of the product, safety precautions, and installation.

Chapter 5 explains information on communication functions such as external control of the MPM-210H.

Chapter 6 explains the trigger functions.

Chapter 7 explains the analog output (MPM-212).

Chapter 8 explains the specifications.

Chapter 9 describes the maintenance and storage methods of the product.

Chapter 10 explains the packing instructions and shipping.

Chapter 11 describes the troubleshooting.

Note:

Before using this product, please carefully read this Operation Manual.

Markings on the instrument



This mark is used on the equipment as warning of danger.

Explanation of Terms

The meanings of the following terms used in this Operation Manual are defined as below:

(1) Meaning



DANGER

This indicates pressing DANGER, and if it is not avoided, personnel death or serious injury may result; therefore, it is the most emphasized special information.



WARNING

This indicates potential danger, and if it is not avoided, personnel death or serious injury may result; therefore, it is special information.



CAUTION

This indicates potential danger, and if it is not avoided, mild or slight injury may result; therefore, it is special information. It also indicates potential danger leading to only physical damage.

NOTE

NOTE

This section only indicates supplementary explanations of the text and thus other than DANGER, WARNING, and CAUTION.

(2) Importance of Information

From the above meanings, the priority of the terms here are as shown below:

DANGER > WARNING > CAUTION > NOTE

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Instrument Markings

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1. Introduction

The MPM-210H is a multi-port optical power meter for measuring optical characteristic of optical devices. The MPM-210H is composed of up to five MPM-211/MPM-212/MPM213/MPM-215 Measurement Modules (MM). There are four types of measurement modules: MPM-211, MPM-212, MPM-213 and MPM-215. The MPM-211 MM has 4 optical power measuring ports (4 channels). The MPM-212 has 2 optical power measuring ports and 2 analog output ports and outputs analog voltage signal proportional the optical power per each port. The MPM-213 has 4 electrical current measuring ports (4 channels), and is used as same as MPM-211. The MPM-215 MM has 4 optical power measuring ports (4 channels) and enables optical power measurement for each port with a high dynamic range of 60 dB or more. However, the MPM-215 cannot be used as same as other modules (MPM-211/212/213) and it does not support the measurement range switching function. The MPM-210H doesn't have front panel nor display window but it can be easily operated by remote control. The MPM-210H is controllable by using GPIB, LAN, RS232 communication and users can interlock the system with other system (santec Tunable Laser, TSL series and Polarization Controller, PCU series) and operate it for measuring PDL and IL. Commands for controlling the MPM-210H are listed and described in section 5.5.

2. Product Composition

This product is composed of the MPM-210H body and accessories.

The following parts are included:

- MPM-210H Main Frame 1
- MPM-211 / MPM-212 / MPM-213 / MPM-215 Module 1 to 5
- Power cord 1
- USB cable 1
- BNC cable 2
- Operation Manual & Software 1
- Inspection report 1

Check to make sure that the product body and the accessories have no scratches or stains. If any item is found to have scratches or stains, please contact us.

3. Panel Descriptions

3.1. Front Panel

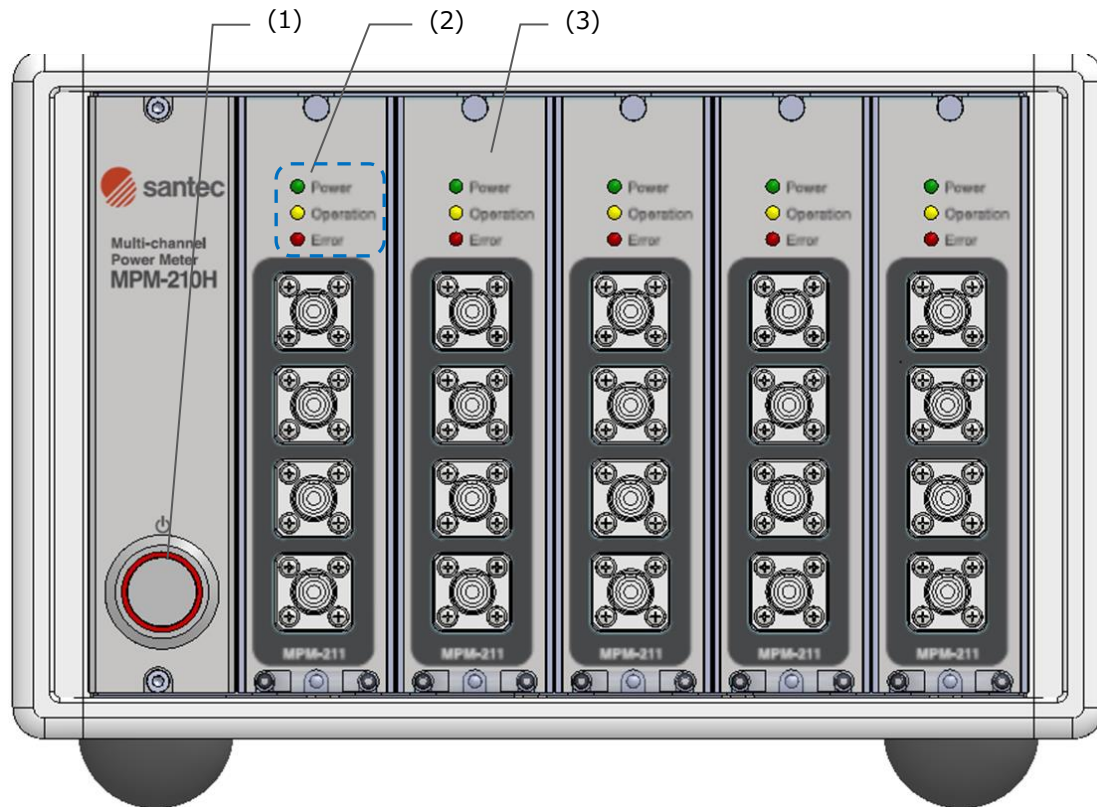


Figure 3-1. Front Panel

- (1) Power on switch
- (2) LED for Power, Operation and Error

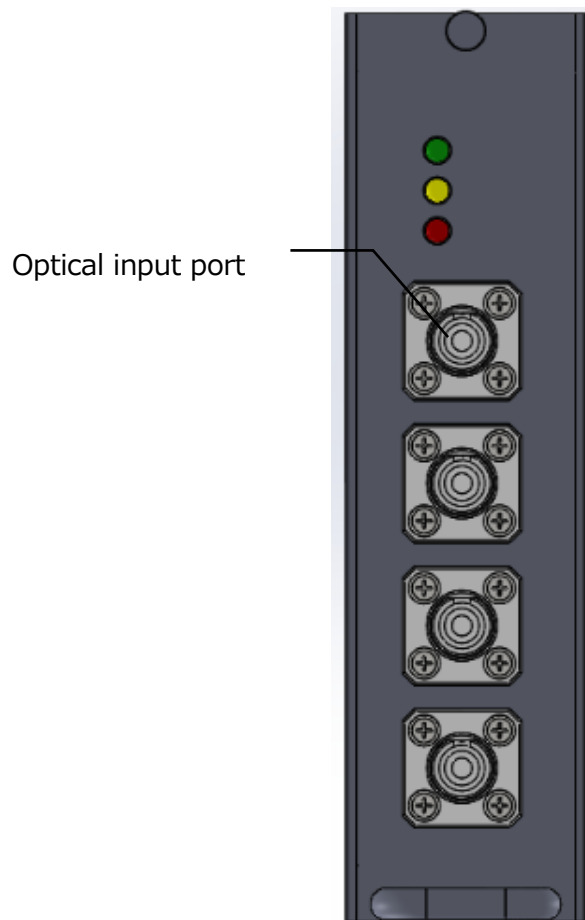
Power LED light will be turned on when the product is on. Operation LED light will be on while measuring and Error LED light will be on if the product is not connected with main body internally. If all the LEDs are lit, the firmware version of Mainframe (MPM-210H) and the firmware version of the modules do not match. Please confirm the firmware version.

- (3) Module

Up to five sensor modules can be inserted, and the module numbers are arranged in order of 0, 1, 2, 3, 4 from the left side.

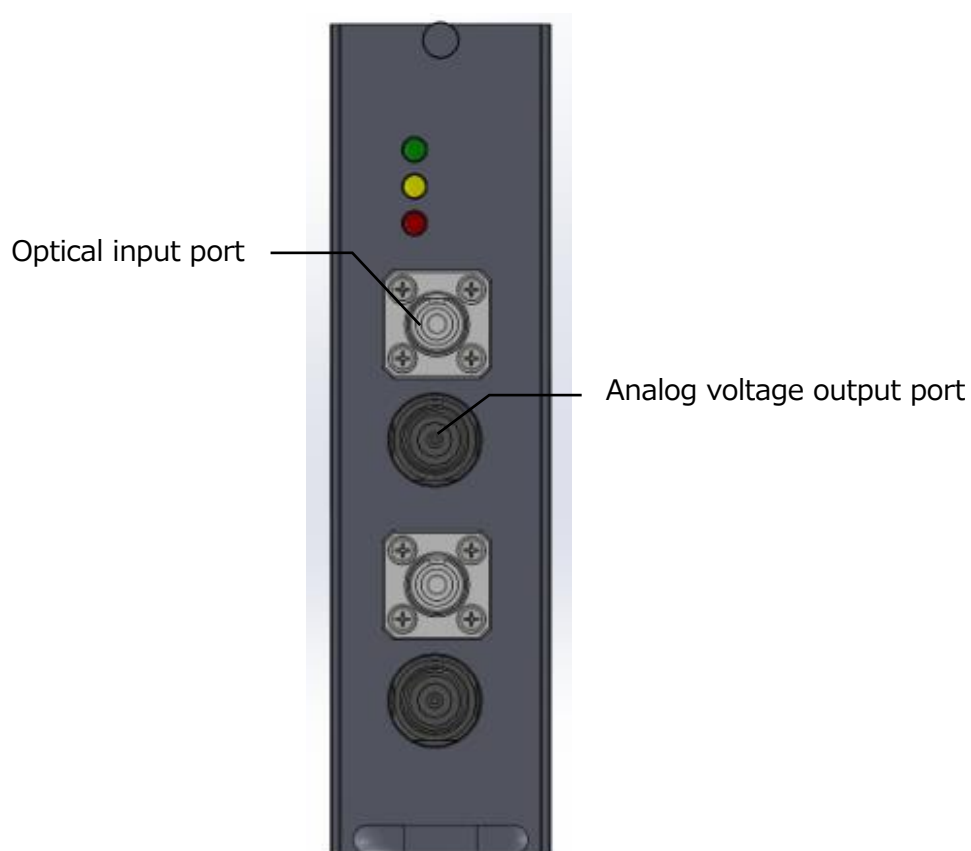
3.2. MPM-211 (Power meter module 4 ports)

The MPM-211 has 4 optical power measuring ports (4 channels). The connector of the optical input port is a FC connector receptacle. Optical input port will be recognized as port for measuring optical power and it gets assigned as port numbers (1,2,3,4) from top to bottom.



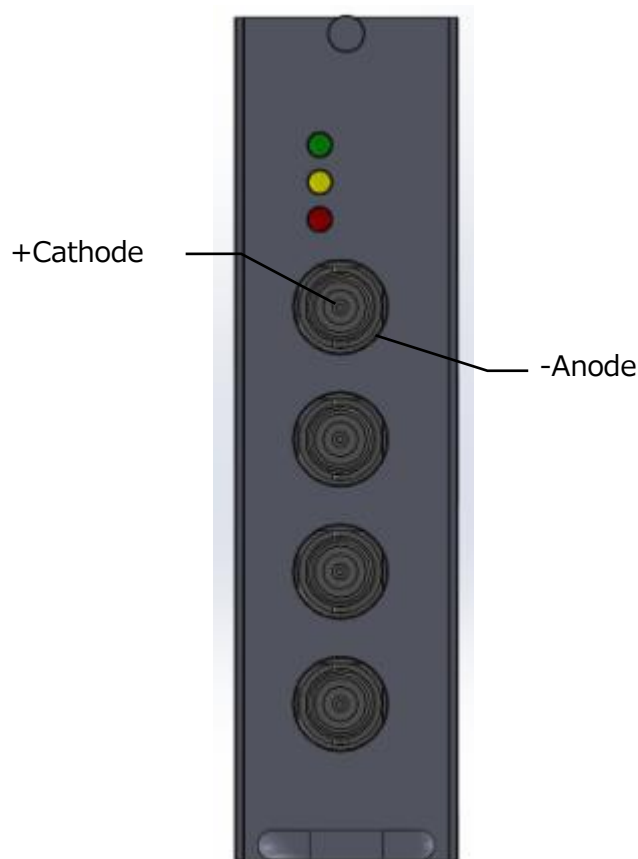
3.3. MPM-212 (Power meter module and analog output 2 ports)

The MPM-212 has 2 optical power measuring ports and 2 analog output ports and outputs analog voltage signal proportional the optical power per each port. The connector of the optical input port is a FC connector receptacle and the analog output port is a BNC male connector. Optical input port 1, analog output port 1, optical input port 2 and analog output port 2 is arranged from top to bottom.



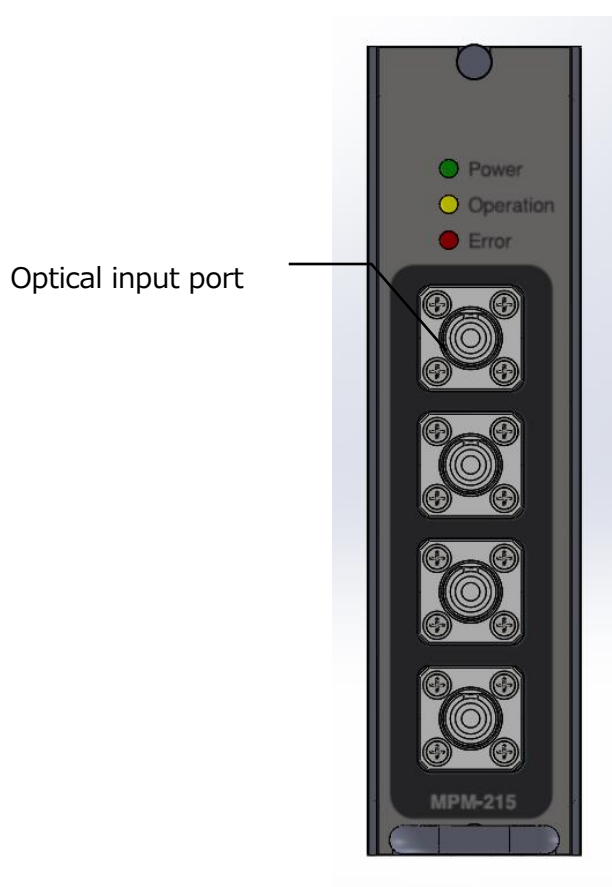
3.4. MPM-213 (Current meter module 4 ports)

The MPM-213 has 4 electrical current measuring ports (4 channels), and is used as same as MPM-211. The connector of the current input port is a BNC male connector. The inside (Signal) of the BNC connector is + Cathode and the outside (Grand) is – Anode. Current input port will be recognized as ports for measuring electrical current and it gets assigned as port numbers (1,2,3,4) from top to bottom. This module is suitable for measuring the performance of fiber optics components using transceiver- like photodiodes (ROSA/Coherent receiver, etc.) and an optical channel monitor.



3.5. MPM-215 (Power meter module 4 ports with high dynamic range)

The MPM-215 has 4 optical power measuring ports (4 channels). The connector of the optical input port is a FC connector receptacle. Optical input port will be recognized as port for measuring optical power and it gets assigned as port numbers (1,2,3,4) from top to bottom. The MPM-215 enables optical power measurement for each port with a high dynamic range of 60 dB or more. However, the MPM-215 cannot be used as same as other modules (MPM-211/212/213) and it does not support the measurement range switching function.



3.6. Rear Panel

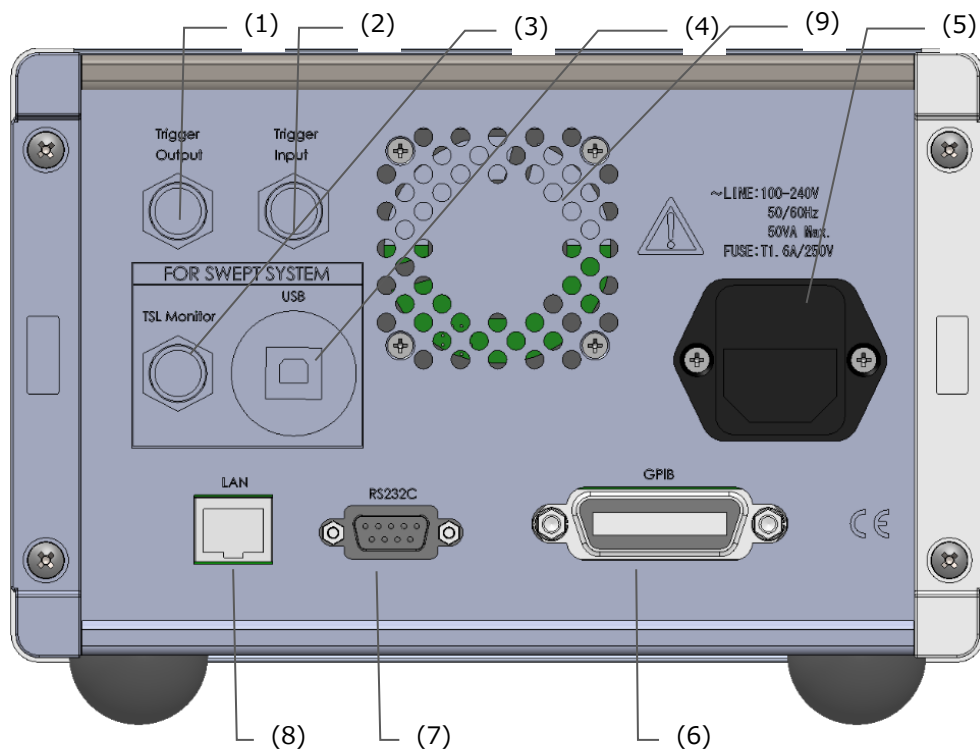


Figure 3-2. Rear Panel

- (1) Trigger Output
- (2) Trigger Input
- (3) TSL Monitor Input (For Swept Test System)
- (4) USB port (For Swept Test System)
- (5) Power Connector Socket and fuse box
- (6) GPIB Port
- (7) RS-232 Port
- (8) LAN Port
- (9) Exhaust fan

4. Installation

4.1. Warning and Caution

- Dangerous voltages, capable of causing injury or death, are present in this instrument. Use extreme caution whenever the instrument covers are removed. Do not remove the covers while the unit is plugged into a live outlet.
- To avoid electric shock, the power cord protective ground conductor must be connected to ground.
- No user serviceable part exists in this instrument. Refer all services to qualified personnel.
- Before using this instrument, be sure to read through operation manual and data sheets to ensure correct operations. Contact SANTEC to obtain service support and do not install substitute parts or perform any unauthorized modifications to this instrument.

4.2. Environment

Pay attention to the following guidelines for safe, trouble-free operation. This product is a high-precision device and be designed to be used indoors (Pollution Degree 2):

- Do not place in direct sunlight.
- Do not place under high temperature and high humidity.
- Do not place in an environment with dust, dirt, salt, or corrosive gas.
- Do not place in an area subject to large vibrations.
- Do not place in an area subject to noise by electric field, magnetic field, etc.
- Do not place in an area where the unit is subject to falling objects landing on it.
- Do not place with possible exposure to water.
- Do not place with an area of altitude 2000m or more.

Clearance of 5 cm at the rear side of the unit should be allowed for proper ventilation. Don't remove rubber feet to ensure air intake.

The mainframe of the MPM-210H weighs about 6.0 kg. Please ensure that the table or shelf upon which it is installed has sufficient strength.

4.3. Line Voltage Selection

This Instrument complies with overvoltage category II and can operate from any single-phase AC power source that supplies 100 - 120V/ 200 - 240V at a frequency at 50/60Hz. The power consumption is 50VA max.

4.4. Line Fuse

The fuse used by this instrument is T1.6A/ 250VAC (100 - 120V/ 200 - 240V).

4.5. Line Cord

- This instrument has three-wire power cord for connection to the power source and to a protective ground.
- The exposed metal chassis of the instrument is connected to ground via the power line cord to protect against electrical shock. Always use a socket outlet that has a properly connected protective ground.



This product is grounded by the ground line in the supplied power cable. This product must be properly grounded to avoid electrical shock and other dangers. Please make sure that this product is properly grounded to avoid electrical shock. In addition, please place this product in a location that allows the user to easily unplug the power cable, in case of any device malfunction.

4.6. Zeroing option (To remove electrical offsets)

The MPM-210H (Optical Power Meters) measures optical power by converting optical power to electrical power of photo detector, and then measuring electrical power. If there is electrical offsets, the MPM-210H may not allow you to measure optical power with accuracy. So, we recommend you performing the zeroing whenever tuning on this instrument or before you need an important measurement. Please refer to "5.5.3. Measurement-related commands" for the detail.

4.7. Maximum input optical power

The maximum input power that can be applied to any port of the power meter is 16dBm. Applying more than the specified power might damage the power meter!

4.8. Inserting a module

The MPM-210H (Main frame) is used to insert some modules (MPM-211/MPM-212/MPM-213/MPM-215). With the power off, please insert the module while paying attention to the orientation of the module. After inserting it, fix it with the knurled screw on the top of the module. The MPM-210H cannot allow you to insert any previous modules (MPM-10/MPM-20).



Insertion of an incompatible module may cause equipment failure. Moreover, please fix the module using knurled screw after inserting each module.

5. Remote Control



Before inserting and extracting the connectors for the communication cables, make sure to turn off power to the device to avoid damaging the device.

The MPM-210H exchanges information with controller by using GPIB, TCPIP or RS232 communication.

5.1. GPIB Communication

The MPM-210H supports IEEE488 interface. When receiving a command, it uses EOI but not EOS. When sending a command, it uses EOI and EOS<CR>.

Default value of GPIB Address is 16.

Connect the 24-pin GPIB cable to the GPIB connector on the rear panel. The total extension of the connected cable shall be 20 m or less. The cable length between each device must be 2 m or less. At most, 15 devices can be connected.



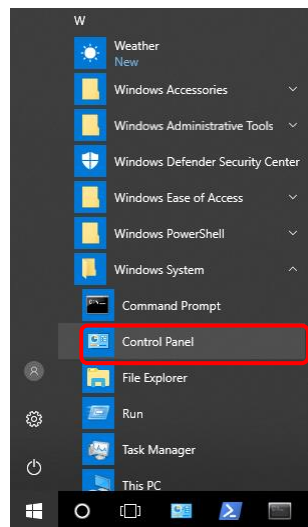
Do not disconnect or connect a cable from and to devices connected with GP-IB cable, do not short-circuit the connector, and do not turn ON/OFF the devices. Otherwise, actions may be stopped or errors may occur, causing a failure. In the event of such failure, reset all the connected devices, and then activate the system once again. When configuring a system, remove the unused device or unnecessary cable and return to the original setup.

5.2. TCP/IP Communication

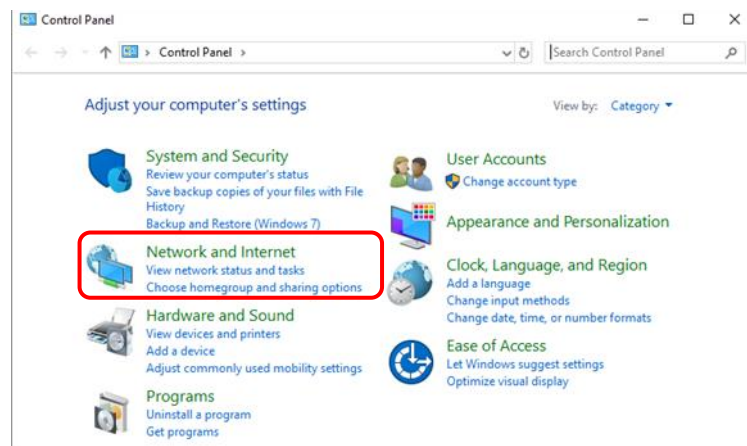
5.2.1 Computer setting method of the TCP/IP communication

In case of the TCP/IP communication, please set up the computer in reference to the following step.

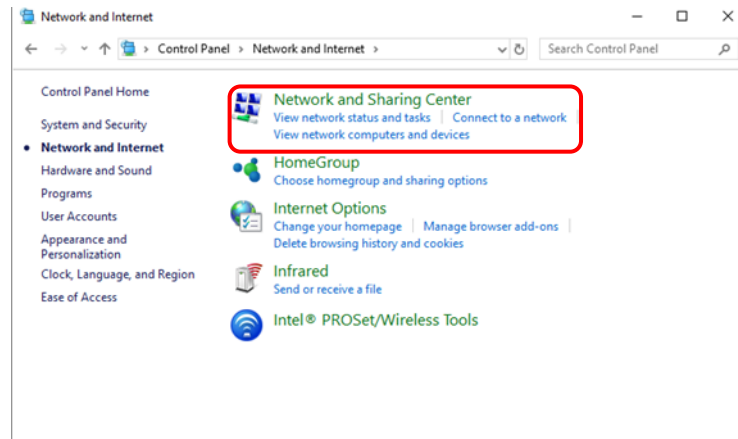
1/Clicks the Start menu and selects the Control Panel.



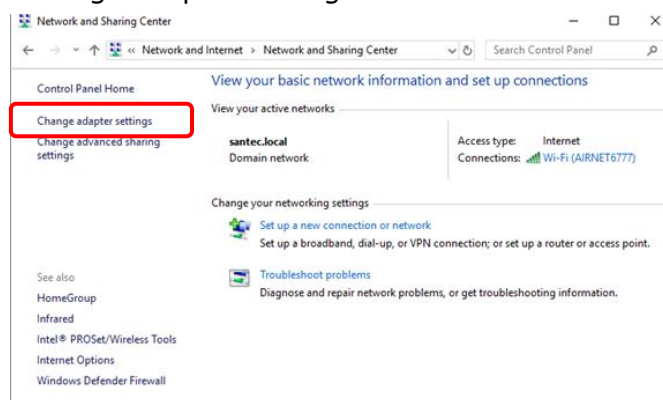
2/Clicks on the “Network and Internet”



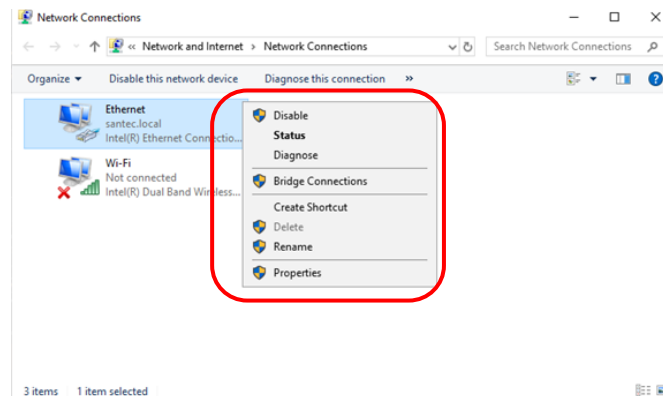
3/Clicks on “Network and Sharing Center”



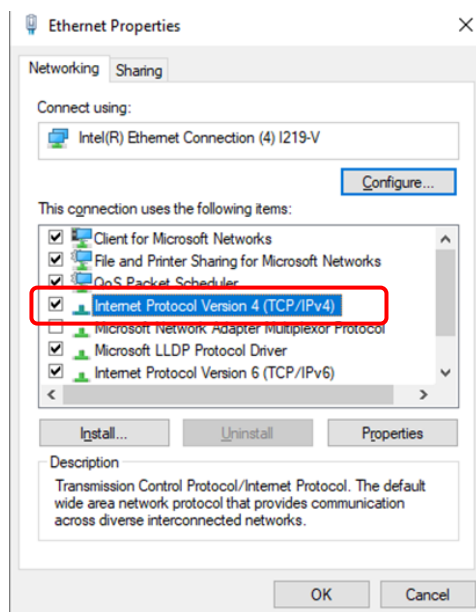
4/Clicks on “Change Adapter Settings”.



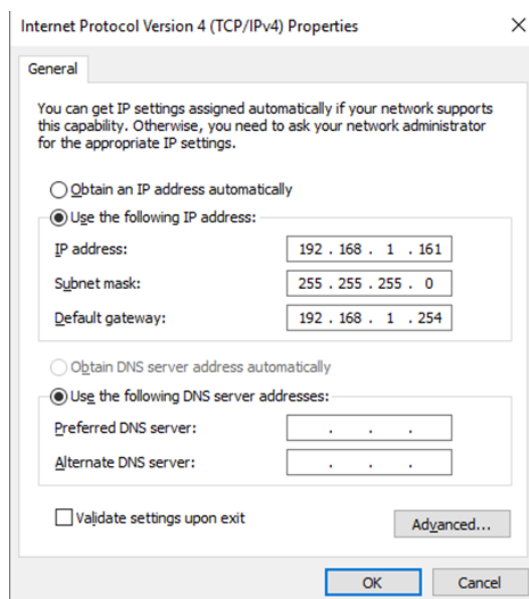
5/Right clicks on “Local Area Connection” and click Properties”.



6/Clicks on “Internet Protocol Version 4 (TCP/IP)” then click “Properties”.



7/Changes to “Use the following IP address” and inputs your IP and DNS information.



Sets the IP address to a different address from the MPM-210H address.

Sets the Subnet mask to same address as the MPM-210H address.

Sets the Default gateway to same address as the MPM-210H address.

*Please check the MPM-210H address by GPIB communication.

Checks the IP address by the command “IP?”.

Checks the Subnet mask by the command “SUBNET?”.

Checks the Default gateway by the command “GW?”.

To set up for Gate way IP, Subnet Mask, IP Address, user needs to set them by RS232 communication or GPIB communication. (Refers to the 5.5.1.)

When the MPM-210H receives the command, the terminator is recognized the <LF> and when the MPM-210H sends the command, the terminator is the <CR><LF>.

* Please use the cable of UL94 V1 or more for TPC / IP communication.

Default Configuration

Item	Command	Default Address
Gate way Address	GW	192.168.1.254
Subnet Mask	SUBNET	255.255.255.0
IP Address	IP	192.168.1.161
Remote port	-	5000

NOTE

When sending each command by the TCP/IP communication, the wait of 10ms needs after sending commands. Please refer to the "5.6.7. Sample code for "SME mode measurement" for the detail.

5.3. RS-232 Communication

5.3.1 RS-232 Communication Set up

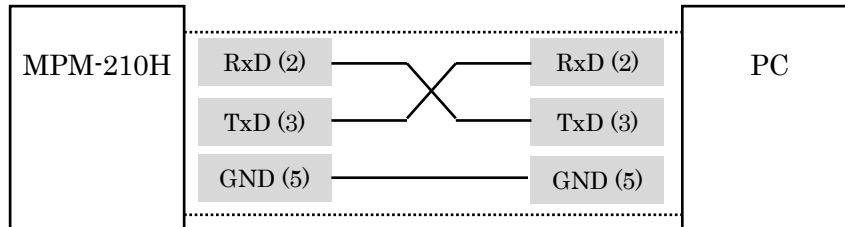
Baud Rate	115200
Data Bits	8 bits
Parity Bit	None
Stop Bit	1
Terminator	LF
Echo	False (Default)

When the echo is valid, sets it by the "ECHO" command. Refer to the section "5.5.2" for the detail. When the MPM-210H receives the command, the terminator is recognized the <LF> and when the MPM-210H sends the command, the terminator is the <CR><LF>.

The Logging function is not available for RS-232 communication.

5.3.2 RS-232 Connection

The PC and the MPM-210H are connected each other by D-SUB 9 pin female serial cable. Methods of Serial cable pin connection is as follows. Uses the cross cable.



5.4. Commands Form

Structures of commands are formed by Commands and Parameter.

Basic structures are as follows.

**COMMAND<wsp> [<Parameter1>][,<Parameter2>][,<Parameter3>]
[] : Optional.**

- COMMAND is not case-sensitive.
- <wsp> stands for the white space.
- There are string type, integer type and real type in Parameter, decimal notation is used for real type.
- When receiving with RS232C and TCP/IP, the end of Commands character string is determined by using ASCII <LF> and <CR> will be ignored.
When receiving with GPIB, the end of Commands character string is determined by EOI and <CR>, <LF> will be ignored.
- Query Command is an answer command. Users have to put '?' (Question mark) at the end all the time.
- If there are multiple parameters that are received as answers for Query Commands, it separated by ','(comma), have <CR><LF> at the end of parameters in case of RS232C and TCP/IP, and have <CR> at the end of parameters in case of GPIB.

No acceptable for COMMAND & Parameter	' % ', ' & ', ' \$ ', ' # ', ' ~ '
Dividing character for COMMAND & Parameter	' , ', ' _ ', ' ' , ' : '

Not acceptable for dividing COMMAND & Parameter	' ` ' , ' ¥t '
---	----------------

- Special characters process as follows (When receiving)

BS(back space)	Delete last received character
<CR>(Carriage Return)	Ignore
<LF>(Line Feed)	Handles it as a Terminating character.

5.5. Commands

5.5.1. Commands list

1) System-related commands

Command	Description	Page
ECHO	Definite echo for RS-232 communication	27
*IDN?	Identification Query	27
ERR?	Check Error information.	28
IDIS?	Check recognition of MM for MPM-210H.	28
MMVER?	Identification query of a module type	28
ADDR/ADDR?	Sets/Reads out the GPIB address.	29
GW/GW?	Sets/Reads out the Gate way Address.	29
SUBNET/SUBNET?	Sets/Reads out the SUBNET Address.	30
IP/IP?	Sets/Reads out the IP Address.	30

2) Measurement-related commands

Command	Description	Page
TRIG/TRIG?	Sets up/Reads out the trigger mode.	30
WMOD/WMOD?	Sets up/Reads out the measurement mode.	31
WAV/WAV?	Sets up/Reads out the measurement wavelength of Constant wavelength Measurement Mode (CONST1, CONST2, READ).	32
DWAV/DWAV?	Sets up/Reads out the measurement wavelength of Constant wavelength Measurement Mode (CONST1, CONST2, READ) for each channels.	33
WSET/WSET?	Sets up /Reads out the measurement wavelength of Sweep Wavelength Measurement Mode (SWEEP1, SWEEP2).	33
SPE/SEP?	Sets up/Reads out the Wavelength Sweep Speed	34
LEV/LEV?	Sets up/Reads out TIA Gain for CONST1, SWEEP1, AUTO1 measuring mode.	35
DLEV/DLEV?	Sets up/Reads out TIA Gain for CONST1, SWEEP1, AUTO1 measuring mode for each channels.	35
AVG/AVG?	Sets up/Reads out Average Time.	36

FGSAVG/FGSAVG?	Sets up/Reads out Average Time. (This command is same operation as "AVG".)	37
UNIT/UNIT?	Sets up/Reads out measuring unit of optical power or electrical current.	38
AUTO/AUTO?	Sets up/Reads out the power range (Auto or Manual).	38
DAUTO/DAUTO?	Sets up/Reads out the power range (Auto or Manual) for each channels.	39
READ?	Execute optical power or electrical current measurement and check the result.	39
CWAV?	Checks the wavelength that should be calibrated.	40
CWAVPO?	Checks Power calibration value of the wavelength of "CWAV?" command Index.	40
MEAS	Command to start measuring.	41
STOP	Command to stop measuring.	41
STAT?	Checks the measuring status.	41
LOGN/LOGN?	Sets up/Reads out measurement data point in CONST1/CONST2/FREERUN measuring mode.	41
LOGG?	Read out the logging data.	42
ZERO	Before measuring optical power, run Zeroing to delete electrical DC offset.	42

5.5.2. Command compatible with the previous MPM-210 model

The MPM-210H is command compatible with the previous MPM-210 model, so switching to the new MPM-210H is as simple as swapping out your devices.

5.5.3. System-related commands

1. ECHO

Description	Definite echo (For only RS-232 Communication)
Syntax	ECHO<wsp><value>
Parameters	0 : Disable 1 : Enable
Default	0
Example	ECHO 0

2. *IDN?

Description	Identification Query
Syntax	*IDN?

Response	SANTEC,MPM-210H,zzzzzzzz,Ver. x.y zzzzzzzz is serial number of main frame, and Ver. x.y is firmware version.
Example	*IDN? Response: SANTEC,MPM-210H,00000000,Ver.2.0

3. ERR?

Description	Check Error information.
Syntax	ERR?
Response	<value>,<string> <value>: Error Code <string>: Summary of content for Error Please refer to "5.5.5 ERR?"
Example	ERR? Response: 0, "No Error"

4. IDIS?

Description	Check recognition of MM (Module) for MPM-210H.
Syntax	IDIS?
Response	<value0>, <value1>, <value2>, <value3>, <value4> value0: Module0 value1: Module1 value2: Module2 value3: Module3 value4: Module4 0: Module is not recognized. 1: Module is recognized.
Example	IDIS? Response: 1,1,1,1,1

5. MMVER ?

Description	Identification query of a module type
Syntax	MMVER?<wsp><module>
Parameters	<module >: 0,1,2,3,4
Response	<value0>,<value1>,<value2>,<Value3>

	value0: Company (Santec) value1: Product code (MPM-211,MPM-212,MPM-213,MPM-215) value2: Serial number value3: Firmware version
Example	MMVER? 0 Response : Santec,MPM-211,00000000M211,Ver1.11

6. ADDR

Description	Set up GPIB Address
Syntax	ADDR<wsp><value>
Parameters	<value>: 1 ~ 31
Default	16
Example	ADDR 16

7. ADDR?

Description	Check GPIB Address.
Syntax	ADDR?
Response	<value>
Example	ADDR? Response: 16

8. GW

Description	Set up Gate way Address.
Syntax	GW<wsp><www.xxx,yyy,zzz>
Parameters	<www.xxx,yyy,zzz>: 0 ~ 255
Default	192.168.1.254
Example	GW 192.168.1.254

9. GW?

Description	Check Gate way Address.
Syntax	GW?
Response	<www.xxx,yyy,zzz>
Example	GW? Response: 192.168.1.254

10. SUBNET

Description	Set up Subnet Mask.
Syntax	SUBNET<wsp><www.xxx,yyy,zzz>
Parameters	<www.xxx,yyy,zzz>: 0 ~ 255
Default	255. 255. 255.0
Example	SUBNET 255. 255. 255.0

11. SUBNET?

Description	Check Subnet Mask
Syntax	SUBNET?
Response	<www.xxx,yyy,zzz>
Example	SUBNET? Response: 255. 255. 255.0

12. IP

Description	Set up IP Address.
Syntax	IP<wsp><www.xxx,yyy,zzz>
Parameters	<www.xxx,yyy,zzz>: 0 ~ 255
Default	192.168.1.161
Example	IP 192.168.1.161

13. IP?

Description	Check IP Address.
Syntax	IP?
Response	<www.xxx,yyy,zzz>
Example	IP? Response: 192.168.1.161

5.5.4. Measurement-related commands

1. TRIG

Description	If measuring mode is 'CONSTx', it checks for measuring trigger.
Syntax	TRIG<wsp><value>
Parameters	<value>

	0 – Internal trigger 1 – External trigger If it is internal trigger, it generates trigger every 10ms.
Default	1
Example	TRIG 1

2. TRIG?

Description	If measuring mode is 'CONSTx', it checks for measuring trigger.
Syntax	TRIG?
Response	<value> 0 – Internal trigger 1 – External trigger
Example	TRIG? 1 Response: 1

3. WMOD

Description	Set measurement mode. The MPM-215 supports only "CONST2" and "SWEEP2".
Syntax	WMOD<wsp><value>
Parameters	<value> "CONST1" => Constant Wavelength, No Auto Gain, SME mode "SWEEP1" => Sweep Wavelength, No Auto Gain, SME mode "CONST2" => Constant Wavelength, Auto Gain, SME mode "SWEEP2" => Sweep Wavelength, Auto Gain, SME mode "FREERUN" => Constant Wavelength, No Auto Gain, First Hardware Trigger Start (CEM mode) Refer to the 5.6.1 WMOD.
Default	CONST1
Example	WMOD CONST1

4. WMOD?

Description	Check measurement mode. The MPM-215 supports only "CONST2" and "SWEEP2".
Syntax	WMOD?
Response	"CONST1" => Constant Wavelength, No Auto Gain, SME mode "SWEEP1" => Sweep Wavelength, No Auto Gain, SME mode "CONST2" => Constant Wavelength, Auto Gain, SME mode "SWEEP2" => Sweep Wavelength, Auto Gain, SME mode "FREERUN" => Constant Wavelength, No Auto Gain, First Hardware Trigger Start (CEM mode) Refer to the 5.6.1 WMOD.
Example	WMOD? Response: CONST1

5. WAV

Description	Set up for using wavelength of Constant wavelength Measurement Mode (CONST1, CONST2, READ).
Syntax	WAV<wsp><value>
Parameters	<value> 1250.000 ~ 1630.000 Real number Unit: nm
Default	1550
Example	WAV 1550

6. WAV?

Description	Check using wavelength of Constant wavelength Measurement Mode (CONST1, CONST2, READ).
Syntax	WAV?
Response	<value> 1250.000 ~ 1630.000 Real number Unit: nm
Example	WAV? Response: 1550

7. DWAV

Description	Set up for using wavelength of Constant wavelength Measurement Mode (CONST1, CONST2, READ) for each channel.
Syntax	DWAV<wsp><value1>,<value2>,<value3>
Parameters	<value1>: Module 0, 1, 2, 3, 4, 5 When setting the <value 1> to 5, all channels and module is set to the same wavelength. <value2>: Channel 1, 2, 3, 4 <value3>: 1250.000 ~ 1630.000 Real number Unit: nm
Default	1550
Example	DWAV 0,1,1550

8. DWAV?

Description	Check using wavelength of Constant wavelength Measurement Mode (CONST1, CONST2, READ) for each channel.
Syntax	DWAV? <wsp><value1>,<value2>
Parameters	<value1>: Module 0,1, 2, 3, 4 <value2>: Channel 1, 2, 3, 4
Response	<value> 1250.000 ~ 1630.000 Real number Unit: nm
Example	DWAV? 0,1 Response: 1550

9. WSET

Description	Set up for using wavelength of Sweep Wavelength Measurement Mode (SWEEP1, SWEEP2). The data points is calculated from the condition that is set by the "WSET" command and it is possible to check the data points by the "LOGN?" command. Please refer to "5.6.3. Measurement data point setting (LOGN)".
-------------	---

Syntax	WSET<wsp><value1>,<value2>,<value3>
Parameters	<value1>: Start wavelength – 1250 ~ 1630 <value2>: Stop wavelength – 1250 ~ 1630 <value3>: Step wavelength – 0.001 ~ 10 Unit: nm Note) Stop wavelength = Start wavelength + step wavelength
Default	value1: 1250, value2: 1630, value3: 0.005
Example	WSET 1520,1570,0.05

10. WSET?

Description	Check using wavelength of Sweep Wavelength Measurement Mode (SWEEP1, SWEEP2).
Syntax	WSET?
Response	<value1>,<value2>,<value3> <value1>: Start wavelength – 1250 ~ 1630 <value2>: Stop wavelength – 1250 ~ 1630 <value3>: Step wavelength – 0.001 ~ 10 Unit: nm
Example	WSET? Response: 1520,1570,0.05

11. SPE

Description	Set up Wavelength Sweep Speed (Sweep Wavelength Measurement Mode (SWEEP1, SWEEP2)). The averaging time is automatically set by the measurement speed and measurement step of WSET.
Syntax	SPE<wsp><value>
Parameters	<value>: Sweep speed – 0.001 ~ 200 Unit: nm/sec
Default	value: 1
Example	SPE 1

12. SPE?

Description	Check Wavelength Sweep Speed (Sweep Wavelength
-------------	--

	Measurement Mode (SWEEP1, SWEEP2)).
Syntax	SPE?
Response	<value> Sweep speed – 0.001 ~ 200 Unit: nm/sec
Example	SPE? Response: 1

13. LEV

Description	Set up TIA Gain for CONST1, SWEEP1, FREERUN, AUTO1 measuring mode. Refer to the 5.5.7 for the relation between the gain setting and the input power. This command is not available for the MPM-215.
Syntax	LEV<wsp><value>
Parameters	<value>: 1, 2, 3, 4, 5 In case of the MPM-213 (current meter), the available range is 1 to 4.
Default	value: 5
Example	LEV 5

14. LEV?

Description	Check TIA Gain of CONST1, SWEEP1, FREERUN, AUTO1 measuring mode. Refer to the 5.5.7 for the relation between the gain setting and the input power. This command is not available for the MPM-215.
Syntax	LEV?
Response	<value> value: 1, 2, 3, 4, 5
Example	LEV? Response: 5

15. DLEV

Description	Set up TIA Gain for CONST1, SWEEP1, FREERUN, AUTO1
-------------	--

	measuring mode for each channel. Refer to the 5.5.7 for the relation between the gain setting and the input power. This command is not available for the MPM-215.
Syntax	DLEV<wsp><value1>,<value2>,<value3>
Parameters	<value1>: Module 0, 1, 2, 3, 4, 5 When setting the <value 1> to 5, all channels and module is set to the same wavelength. <value2>: Channel 1, 2, 3, 4 <value3>: Gain 1, 2, 3, 4, 5 In case of the MPM-213 (current meter), the available range is 1 to 4.
Default	value3: 5
Example	DLEV 0,1,5

16. DLEV?

Description	Check TIA Gain of CONST1, SWEEP1, FREERUN, AUTO1 measuring mode for each channel. Refer to the 5.5.7 for the relation between the gain setting and the input power. When using the "READ?" command, the manual mode (AUTO1) of the range mode is only available. This command is not available for the MPM-215.
Syntax	DLEV? <wsp><value1>,<value2>
Parameters	<value1>: Module 0, 1, 2, 3, 4 <value2>: Channel 1, 2, 3, 4
Response	<value> value: 1, 2, 3, 4, 5
Example	DLEV? 0,1 Response: 5

17. AVG

Description	Set up Average Time.
Syntax	AVG<wsp><value>
Parameters	<value>: 0.01 ~ 10000.00

	Unit: ms Resolution: 0.01(10us)
Default	value: 5
Example	AVG 5

18. AVG?

Description	Check Average time.
Syntax	AVG?
Response	<value> value: 0.01 ~ 10000.00 Unit: ms resolution: 0.01(10us)
Example	AVG? Response: 5

19. FGSavg

Description	Set Average Time. (This command is same operation as "AVG".)
Syntax	FGSAVG<wsp><value>
Parameters	<value>: 0.01 ~ 10000.00 Unit: ms Resolution: 0.01(10us)
Default	value: 0.1
Example	FGSAVG 0.1

20. FGSavg?

Description	Check Average time. (This command is same operation as "AVG?".)
Syntax	FGSAVG?
Response	<value> value: 0.01 ~ 10000.00 Unit: ms resolution: 0.01(10us)
Example	FGSAVG? Response: 0.1

21. UNIT

Description	Set measuring unit of optical power or electrical current.
Syntax	UNIT<wsp><value>
	<value>
Parameters	0 – dBm, dBmA 1 – mW, mA
Default	value: 0
Example	UNIT 0

22. UNIT?

Description	Check measuring unit of optical power or electrical current.
Syntax	UNIT?
	<value>
Response	0 – dBm, dBmA 1 – mW, mA
Example	UNIT? Response: 0

23. AUTO

Description	Set the power range (Auto or Manual). It is available for the "READ?" command.
Syntax	AUTO<wsp><value>
	<value>
Parameters	0 – Manual range 1 – Auto range
Default	value: 1
Example	AUTO 1

24. AUTO?

Description	Check the power range (Auto or Manual). It is available for the "READ?" command.
Syntax	AUTO?
Response	0 – Manual range 1 – Auto range

Example	AUTO? Response: 1
---------	----------------------

25. DAUTO

Description	Set the power range (Auto or Manual) for each channels. It is available for the "READ?" command.
Syntax	DAUTO<wsp><value1>,<value2>
Parameters	<value1>: Module 0,1, 2, 3, 4, 5 When setting the <value 1> to 5, all channels and module is set to the same wavelength. <value2>: Range mode 0 – Manual range 1 – Auto range
Default	value2: 1
Example	DAUTO 0,1

26. DAUTO?

Description	Check the power range (Auto or Manual) for each channels. It is available for the "READ?" command.
Syntax	DAUTO?<wsp><value>
Response	<value>: Range mode 0 – Manual range 1 – Auto range
Example	DAUTO? 0 Response : 1

27. READ?

Description	Execute optical power or electrical current measurement and check the result. If sending the "READ" command during logging mode, logging mode stops.
Syntax	READ?<wsp><module>
Parameter	<module> Module Number: 0,1,2,3,4
Response	<value1>,<value2>,<value3>,<value4>

	value1; Optical power of port 1 value2; Optical power of port 2 value3; Optical power of port 3 value4; Optical power of port 4
Example	READ? Response: -20.123,-20.454,-20.764,-20.644

28. CWAV?

Description	Check the wavelength that should be calibrated. Unit: nm Refer to the 5.5.6 for the method of wavelength compensation.
Syntax	CWAV?<wsp><module>,<index>
Parameter	<module> Module Number: 0,1,2,3,4 <index> Index : Wavelength set order that would be applied – Integer 1,2,3~ 18,19,20
Response	<value> Wavelength Unit: nm
Example	CWAV? 0,1 Response: 1250

29. CWAVPO?

Description	Power calibration value of the wavelength of "CWAV?" command Index. Unit: dB Refer to the 5.5.6 for the method of wavelength compensation.
Syntax	CWAVPO?<wsp><module>,<ch>,<index>
Parameter	<module> Module Number: 0,1,2,3,4 - Integer <ch> Port number: 1,2,3,4 - Integer <index>

	Index : Wavelength set order that would be applied – Integer 1,2,3~ 18,19
Response	<optical power offset> - float
Example	CWAVPO? 0,1,1 Response: 0.904640

30. MEAS

Description	Command to start measuring.
Syntax	MEAS
Example	MEAS

31. STOP

Description	Command to stop measuring.
Syntax	STOP
Example	STOP

32. STAT?

Description	Check measuring status.
Syntax	STAT?
Response	<value1>,<value2> <value1>: Status 0 – Measuring is still in process. 1 – Measurement completed. -1 – The measurement is forcibly stopped. <value2>: Measured logging point
Example	STAT? Response: 1,100

33. LOGN

Description	Set up measurement data point in CONST1/CONST2/FREERUN measuring mode. Refer to the 5.6.3 Measurement data setting (LOGN).
Syntax	LOGN<wsp><value>
Parameters	<value>: 1 ~ 1,000,000
Default	value: 1

Example	LOGN 100
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34. LOGN?

Description	Check measurement data point. Refer to the 5.6.3 Measurement data setting (LOGN).
Syntax	LOGN?
Response	<value>: 1 ~ 1,000,000
Example	LOGN? Response: 100

35. LOGG?

Description	Read out the logging data. This command is not available for RS-232 communication.
Syntax	LOGG? <wsp><value1>,<value2> <value1>: module number <value2>: port number
Response	Please refer to '5.6.4 LOGG?' for detail information.
Example	LOGG? Response: Please refer to '5.6.4 LOGG?' for detail information.

36. ZERO

Description	Before measuring optical power, run Zeroing to delete electrical DC offset. Please be careful with incidence light into Optic Port. When using the current meter module, the MPM-213, please remove the BNC cable from the MPM-213. This command action takes about 3 sec so please run other commands at least 3 sec later.
Syntax	ZERO
Example	ZERO

5.5.5 ERR?

All kinds of errors will be saved into message Que. By using a command 'ERR?', users can pull out error information and read them all.

Number	String
0	No error
-101	"Invalid character" This occurs when unacceptable characters are received for Command or Parameter. Unacceptable characters: "%", "&", "\$", "#", "~"
-103	"Invalid separator" This occurs when an unacceptable character is received as a separator between the Command and the Parameter Unacceptable characters: "`", ":", "
-104	"Data type error" This occurs when the Parameter is not an acceptable data type.
-108	"Parameter not allowed" This occurs when the number of parameters in the corresponding command is more or less
-109	"Missing parameter" This occurs when the number of characters in the Parameter is longer than 18.
-110	"Command header error" This occurs when the number of characters in the Command is longer than 13.
-113	"Undefined Header" This occurs when an unsupported command is received.
-221	"Setting conflict" This occurs when one of the following set up commands, other than "STOP" or "STAT?", was received before measurement using "MEAS" command is completed.. Commands: AVG, LEV, LOGN, AUTO, WAVE, WMOD, WSET, SPE, LOOP, UNIT
-222	"Data out of range" This occurs when the parameter is outside the acceptable value.
-284	"Program currently running" This occurs when the mainframe delivers new commands to the module before the process of delivering commands to the module and receiving responses is completed.
-300	"Device specific error" This occurs when the GPIB Address number that you are trying to set exceeds 32.
-301	"Is not Measurement Module" This occurs when user attempts to deliver a command to a module (slot) that is not installed.
-350	"Queue overflow" This occurs 1. when the Queue space which is used to communicate between internal Tasks is full of messages and there is no space to store any information. 2. when there is no Queue space to store generated Error information
-351	"Queue empty" This occurs when there is no following message in the Queue space which is used

	between internal Tasks to communicate information.
101	"uPP Comm. Header Error" This occurs when the Headers of the Packet which is used to send and receive data between the mainframe and the module are different.
103	"uPP Comm. Rsp No" This occurs when the mainframe sends data information to the module but does not receive a response
104	"uPP Comm. Module Mismatched" This occurs when the mainframe receives information from the other modules than the module that sent data information to.
110	"TCP/IP Comm. Error" This occurs when all data to be transferred is not sent in TCP/IP communication.
116	"GPIB Tx not completed" This occurs when no event is delivered to the internal GPIB Task used for GPIB communication.
117	"GPIB Tx Timer Expired" This occurs when all data to be transferred is not sent in GPIB communication.
120	"MC Trig. Error" This occurs when the mainframe does not receive a measurement completion signal (H/W signal) from the module after the measurement command was delivered to the module.
210	"not exist SEM" This occurs when unregistered message is delivered between internal tasks.

5.5.6 A method for wavelength compensation

The MPM-210H has wavelength calibration values of each port. Number of calibrated wavelength is 20. (From 1250nm to 1630nm with 20nm space). Wavelength between each calibrated wavelength is calculated using 1'st Order linear interpolation. You can check calibrated wavelength with "CWAV?" command. Calibrated values of 20 wavelengths can be checked also with "CWAVPO?" command.

Calibration Wavelength: 1250, 1270, ~, 1630 nm

Interval: 20nm

Calibration Optical power Offset Unit: dB

1 Order Linear Interpolation

$$y = (y_1 - y_0) / (x_1 - x_0) * (x - x_0) + y_0$$

x: wavelength, y: optical power offset

Run measurement with fixed wavelength in FREERUN measurement mode.

Optical power value that was measured with the MPM-210H executes calibration for the fixed wavelength. To do calibration for other wavelengths, refer to following formula.

FP_n : *Optic Power –FREERUN Mode*

$n = 1, 2, \dots, N$ N : *Logging Number*

WOP_n : *Wavelength Optic Power Offset*

$FWOP$: *fixed Wavelength Optic Power Offset*

$$CP_n = FP_n + WOP_n - FWOP$$

WOP_n is calculated with 1 Order Linear Interpolation after getting data using "CWAV?" and "CWAVPO?".

$FWOP$ is the calibrated value of wavelength that was set using "WAV". We recommend to set wavelength as 1550nm. Set wavelength of equipment that are used for calibrating MPM-210H is 1550nm. Optical power of wavelength that is set using "WAV" does not need to be wavelength calibration.

5.5.7 Measurement range (LEV/LEV?/DLEV/DLEV?)

In case of the measurement mode CONST1, SWEEP1 and FREERUN, the "LEV" or "DLEV" command sets the power range and the "LEV?" or "DLEV?" command reads out the set power range. The Table 5.5.1 shows the corresponding power for each range. The Upper Power Limit is always 2dB greater than the optical power value of the chosen range. However, the measurement performance of larger power than the following power for each Range is out of warranty.

Table 5.5.1 Range vs. input power

Range	Optical power (dBm) or current (dBmA)		
	MPM-211/212	MPM-213	MPM-215
1	<+10	<+10	<+5
2	<+0	<+0	-
3	<-10	<-10	-
4	<-20	<-20	-
5	<-30	-	-

NOTE

It is not available for the MPM-215. When using the MPM-215, the measurement range cannot be switched, but the MPM-215 can measure the optical power for each port with a high dynamic range larger than 60 dB without switching the measurement range. When high dynamic range measurement is required, it is recommended to input the highest power possible (The maximum input power to the MPM-215 is +5dBm).

5.6. Logging mode

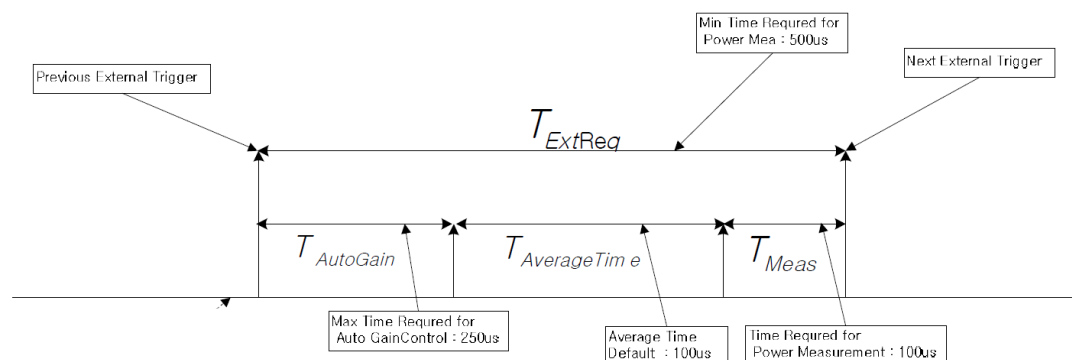
5.6.1. Measurement mode (WMOD)

The MPM-210H has "CONST1", "SWEEP1", "CONST2", "SWEEP2" and "FREERUN" as the measurement mode. Each mode function shows as below.

- CONST1, CONST2: Measure optical power at the fixed wavelength.
- SWEEP1, SWEEP2: Measure optical power within the set wavelength range that is set by the "WSET" command.
- CONST1, SWEEP1, FREERUN: Measure optical power at the fixed range that is set by the "LEV" command. These mode give priority to the measurement accuracy.
- CONST2, SWEEP2: Measure optical power at the auto range. These mode give priority to the measurement speed.
- FREERUN: Measure optical power every the set averaging time. The measurement starts from the hardware trigger.

5.6.2. Fast Gain Switching Average Time

When the SWEEP measurement mode (SWEEP 2) is used with auto gain, please pay attention to the timing of input trigger. The drawing at the bottom explains about measuring time and gap between triggers on SWEEP2 measuring mode.



SWEEP2 measurement time chart

External trigger should be entered as longer than T_{ExtReq} , otherwise external trigger will be ignored. To have more accurate measuring result, not focusing on speed of measurement, users can increase $T_{AverageTime}$. When $T_{AverageTime}$ is increased, T_{ExtReq} also increases and it results as gap between external triggers get increased. When users changes $T_{AverageTime}$, either command "FGSAVG" or "AVG" can be supported.

This section does not apply to the MPM-215 since there is no switching of measurement range ("5.5.7").

5.6.3. Measurement data point setting (LOGN)

In case of the SWEEP measurement mode (SWEEP1, SWEEP2), the measurement data point is calibrated using the set value by the "WSET" command.

Data point = (Stop wavelength – Start wavelength) / Wavelength step + 1

This number can be queried with the LOGN? Command.

In case of the CONST1, CONST2 and FREERUN measurement mode, users set the measurement data point by the "LOGN" command.

5.6.4. LOGG?

If the installed module is MRM-211/MPM-212/MPM-215, it acquires optical power. In case of the MPM-213, it acquires electrical current.

This command is not available for RS-232 communication.

Trans. Order	Data	Data Format	Description
1	'#'	ASCII	Send '#'
2	'Number of digits'	ASCII	Send 'the number of digit,' namely, the number of data. (Note: Here, 'the number of digits' is indicated with a one digit number and means the number of digits when the total number of data transmitted is converted into bytes.)
3	'Number of data'	ASCII	Send 'the number of data,' namely, the number of data to be transmitted. (Note: Here, 'the number of data' means the number obtained by converting the total number of data into bytes. A data is converted into 4

			bytes.)
4	Measuring data	BINARY	Send measuring data. (Note: Send 4-byte data through 4 times and 1 byte at each time by dividing a number in floating point format as above. Send 4-byte data in order of LSB to MSB four times and 1 byte at each time.)

Example 1) Sending 100 measuring data

In case 100 numbers in floating point format are sent, single precision floating point data is transferred as four bytes data every on byte. If the whole data is converted into bytes, it is 400 bytes long.

Table 5.6.1 Example of binary data transmission

Order of transmission	1	2	3	4	5	6	7	8	...	405
Data format	ASCII	ASCII	ASCII	ASCII	ASCII	BINARY	BINARY	BINARY	...	BINARY
Receiving	#	3	4	0	0	1byte	1byte	1byte	...	1byte

Table 5.6.1 shows the order of transmission when MPM-210H sends 100 measuring data to PC. In GPIB and TCP/IP, communication data is sent in the same way as in Table 5.6.1.

Binary data format used in MPM-210H is IEEE Standard Floating Point Format as in Table 5.6.2. (Note: Binary data is sent every 8 bit from least significant byte to most significant byte.)

Table 5.6.2 Floating Point Data Structure

Name	Sign bit	Exponent bit	Fraction bit
Symbol	S	E	F
Bit	31(MSB)	30 ~ 23	22 ~ 0(LSB)

Sign bit is the 31st bit. '0' means a positive number and '1' means negative number. Exponent is composed of 8 bits from the 30th bit to the 23rd one, and it can express a number between 0~255. The median of Exponents is 127. If Exponent is larger than 127 it is positive, and if it is smaller than 127 it is negative. That is, 127 means 2^0 , 128 2^1 and 126 2^{-1} . Fraction indicates a decimal below 1. The 22nd bit is the decimal number of 1/2, the 21st bit the

decimal number of 1/4, and the 20th bit the decimal number of 1/8. Each of the lower bits expresses its corresponding number in the same rule. Based on this structure, binary floating point data are converted into decimal numbers as in Table 5.6.3.

Table 5.6.3 How to convert binary floating point data into decimal numbers

Description	Sign bit	Exponent bit	Fraction bit
Decimal Value	$(-1)^S \times 2^{E-127} \times (1 + F \times 2^{-23})$		
NaN	Don't Care	255	$\neq 0$
-Infinite	1	255	0
+Infinite	0	255	0

Represents the transformation example 4 byte binary data as below.

Example)

Hexadecimal 4 Byte data

Data	4Byte data				Description
Measurement data	C7	D7	9D	BE	IEEE Standard Floating Point Format
	1 st Byte	2 nd Byte	3 rd Byte	4 th Byte	
Converted data from original data	BE	9D	D7	C7	Original 4 byte data is separated every 8 bit. Each 8 bit rearranges from the last 8 bit data to the first 8 bit data
	4 th Byte	3 rd Byte	2 nd Byte	1 st Byte	

Conversion Hexadecimal (4 Byte data) into Binary

<u>1 0111110</u>	<u>1 0011101</u>	<u>11010111</u>	<u>11000100</u>
4 th Byte	3 rd Byte	2 nd Byte	1 st Byte
Sign(s)	Exponent(E)	Fraction (F)	

Conversion Binary into Decimal

Sign(s)	Exponent(E)	Fraction (F)
1	125	1955780

Conversion into Fractional value

Fractional value	$(-1)^S \times 2^{E-127} \times (1 + F \times 2^{-23})$ $= -1 \times 2^{125-127} \times (1 + 1955780 \times 2^{-23})$ $= -1 \times 0.25 \times 1.233147144317626953125$ $= -0.30828678607940673828125$
------------------	---

- 8. LEV 1 - Set TIA Gain Level.
- 9. MEAS - Time of measurement.
- 10. "Delay 20sec + alpha(3 sec)":
After External Trigger Input "Delay as expected measurement time"
- A. STAT? - Check current measurement status
- 11. STOP - Stop measurement (Option)
- 12. LOGG? 1,1 - Get measured value
- A. Module - 1, port - 1
- 13. CWAV? - Check the wavelength that should be calibrated.
- 14. CWAVPO? - Check the Optical power offset for each calibrated wavelength.
- Executes wavelength sensitivity calibration using the data that are read out from "CWAV?" and "CWAVPO?" command. (Refer to the 5.5.5.)

5.6.7. Sample code for SME mode measurement

- 1. WMOD SWEEP1 - Choose measurement mode
- 2. WSET 1260,1290,0.01 - Set Sweep Wavelength
- 3. SPE 10 - Set Sweep Speed
- 4. LEV 1 - Set TIA Gain Level.
- 5. TRIG 1 - Choose external trigger.
- 6. MEAS - Start measurement
- 7. STAT? - Check current measurement status
- 8. STOP - Stop measurement (Option)
- 9. LOGG? 1,1 - Get measured value

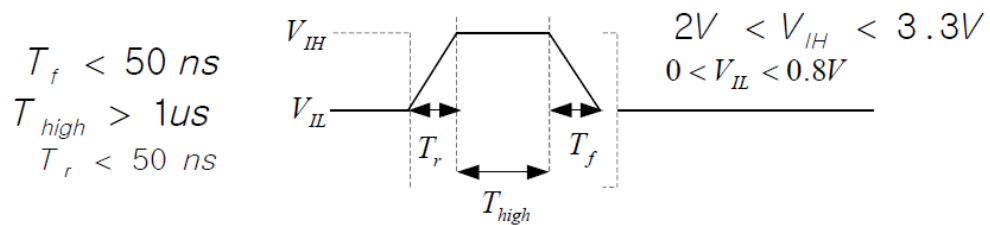
NOTE

In Sweep 1 mode, user does not set Average Time. It is calculated from Sweep speed value and Set wavelength value and applied automatically.

6. Trigger Input/Output

6.1. Trigger Input

Next picture explains about electrical characteristic of External Trigger Signal that enters into Trigger Input Port.



Measurement External Trigger Timing

6.2. Trigger Output

Trigger Output Port outputs each trigger after about 22nsec since External Trigger Signal inputs. Output signal will have electrical characteristic as 3.3v for High and 0v for Low.

7. Analog Output (MPM-212)

The MPM-212 has two analog output ports.

The analog output function outputs a 0 to 2V analog voltage signal proportional to the optical power. When there is no optical input, the analog output function outputs 0V, and when there is maximum optical input, the output voltage is 2V. We recommend using the manual range.

8. Specifications

8.1. Specifications

Table 1: MPM-211/MPM-212/MPM-215 specifications

Parameter	Unit	MPM-211	MPM-212	MPM-215 *1	Notes
Sensor element	-	InGaAs			
Wavelength range	nm	1250 to 1680			
Specification wavelength range	nm	1250 to 1630			
Power dynamic range	dBm	-80 to +10		-70 to +5	
Dynamic range @Fixed gain (typ.) *2, 3	dB	45		70	Logging mode
Number of gain range	-	5		1	
Maximum safe power	dBm	+16			
Total uncertainty	%	±1.5 @ -60 to 9 dBm		±1.5 @ > -55 dBm	
Power resolution	dB	0.001			
Linearity *2, 4	dB	±0.03 @ -55 to 9 dBm		±0.02 @ > -40 dBm ±0.05 @ > -50 dBm	
Polarization dependent responsivity (typ.) *2, 5	dB	< 0.025 < 0.03			@1525-1585 nm @1270-1630 nm
Averaging time	sec	10μ to 10			
Data logging capability	-	1,000,000 x 2buffer per port			
Port number per module	-	4 Ports	2 Ports	4 Ports	
Analog output *6	-	None	With	None	
Connector type	-	FC			

Table 2: MPM-213 specifications

Parameter	Unit	Specifications	Notes
Current dynamic range	dBmA	-70 to +10 @ 100pA to 10 mA	
Dynamic range @Fixed gain (typ.) *2, 3	dB	45 dB	Logging mode
Number of gain range	-	4	
Maximum safe power	dBmA	16	
Total uncertainty	%	±1.5 @ -45 to 9 dBmA ±1 @ > -35 dBm (typ.)	Averaging time > 10ms Averaging time > 10ms
Power resolution	dB	0.001	
Linearity	dB	±0.03 @ -45 to 9 dBmA	Averaging time > 10ms
Averaging time	sec	10μ to 10	
Data logging capability	-	1,000,000 per port x2	
Port number per module	-	4 ports	
Connector type	-	BNC Connector	
Reverse Bias Voltage	V	-	

*1: 1: This module is not compatible with other modules on a same Main Frame. *2: Temperature: 23±5 °C.

*3: Averaging time >50us, Within ±1 °C after zeroing. At gain 1, 2 or 3 for MPM-211/212

*4: For MPM-211/212, Averaging time >100ms, Auto range mode, For MPM-215, Averaging time >30us

*5: SMF, Straight connector *6: Output voltage 0 to 2V, Measurement dynamic range >30dB (typ.)

Table 3: MPM-210H specifications

Parameter	Unit	Specifications	Notes
Module number	-	Up to 5	
Interface	For Power meter For System	GP-IB, Ethernet, RS-232C USB	
Trigger input	-	TTL(3.3 V)	BNC
Trigger output	-	TTL(3.3 V)	BNC
Power monitor	V	0 to 3	BNC
Supply voltage	V	AC 100 to 240, 50/60 Hz	
Maximum power consumption	VA	50	
Operating temperature	°C	10 to 40	
Operating humidity	%	< 80	non condensing
Weight	kg	6	
Dimensions (W) x (D) x (H)	mm	210 x 350 x 133	

Table 4: Common information

Parameter	Unit	Specifications	Notes
Operating temperature	°C	10 to 40	
Operating humidity	%	< 80	non condensing
Dimensions (W) x (D) x (H)	mm	30.3 x 183.5 x 114.8	Connectors are not included.

8.2. Regulations conformity

This product conforms the following standards.

- EMC:

EN 61326-1:2013

EN55011:2009 + A1:2010 (Class A, Group 1)

EN61000-4-2:2009

EN61000-4-3:2006 + A1:2008 + A2:2010

EN61000-4-4:2012

EN61000-4-5:2014 + A1:2017

EN61000-4-6:2014

EN61000-4-8:2010

EN61000-4-11:2004 + A1:2017

FCC 47 CFR Part15 Subpart B Class A (ANSI C63.4:2014)

ICES-003 Issue 7(October 2020) Class A

ICES-003 Issue 6 Class A (ANZI C63.4:2014)

- Safety:

IEC 61010-1: 2010 +

AMD1:2016 IEC 60825-1: 2014(3rd Edition)

- RoHS:

EN 50581:2012

9. Maintenance

9.1 Daily Maintenance

Turn off the power and pull out the power cord before cleaning. Use a soft dry cloth to gently dust the instrument. Do not use chemicals (acetone, alcohol, cleaning fluid, etc.), as they may cause damage to the coating. Do not vigorously rub the display, as damage may result.

9.2 Cleaning the Optical Power Sensor

Dust and dirt on the optical sensor(s) can cause degradation of power measurement performance. To clean the optical power sensor, use an air duster in order to remove dust. We strongly recommend to put optical adapter caps to avoid dust when optical fiber cables are not connected.

9.3 Replacing Fuses

Use two surge resistant type fuses. T1.6A/ 250VAC. Fuses are in the fuse box above the power inlet on the rear panel. Pull out the fuse box, and replace with new fuses.

9.4 Long-Term Storage

■ **Before storing the instrument:**

- 1) Wipe off dust, fingerprints, dirt, stains, etc., from the instrument.
- 2) Use the packing materials that were originally used to package the instrument.
- 3) Avoid storing the instrument in the following types of places.
 - Locations that receive direct sunlight
 - Locations subject to high temperature or high humidity
 - Locations where there's a large amount of dirt, dust, salt content, or corrosive gas
 - Locations subject to vibration
 - Locations where the instrument can be affected by noise from an electric or magnetic field,

10. Re-packing and shipping

Special attention for re-packing and shipping is required when you ship the instrument for repair or to a remote location.

10.1. Re-packing

Please make sure that the original shipping box is used during the shipment, according to the following instructions. If the original shipping box is misplaced or damaged, please contact our customer service department. We can arrange for the shipping box to be sent to you.

Alternately, when the original shipping box is not available, send the instrument with a sturdy, robust shipping box and at least 10cm packing material between the instrument and the shipping box. The entire 10cm space on every side of the instrument should be filled with a cushioning material.



The packing materials we used are designed to protect the instrument from the shock or vibration during the shipment. If the instrument is shipped without packing materials or improperly packed, the instrument might be damaged. Any instrument damage that occurred when the instrument shipped without the original shipping box or without proper packing will not be applicable for warranty repair, even during the guaranteed term.

10.2. Shipping

Treat the box carefully, and avoid shock or vibration as much as possible. Maintain the box in an upright position and label "THIS SIDE UP" in several places.

11. Troubleshooting

Fault condition	Cause	Action
No power	Cord is unconnected.	Connect cord properly.
	No fuse or fuse is blown out	Open the fuse box lid and replace fuse.
GP-IB communication is not possible	GP-IB address is not set correctly	Set GPIB address correctly. (see section 5.2 and 5.5.1)
	Cable is not connected properly.	Check the connection of the cable. (see section 5.2)
All LEDs continue to light even after 30 seconds have passed since the power was turned on.	When any commands are sent to the product during initialization	Please restart the product.
	Mainframe (MPM-210H) and module (MPM-211/212/213/215) firmware version mismatch	Update the firmware.

If after checking the above list, and the device is still not operating correctly, please contact us.



In the event of any trouble with this product, turn the unit off in accordance with the procedures to shut off the power described in this operation manual, disconnect the power source cord, record the product name and serial number described on the name plate of the product, and then contact our dealer at your place or directly contact us at Santec Photonics Laboratories. Our telephone number and facsimile number are shown below. However, we are not responsible for any trouble arising from your own repair or modification on this product.

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